Executive Summary: Hospital Operations & Financial Intelligence

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Purpose

This summary reports what a four-page Power BI dashboard shows about one hospital's activity, costs and payer coverage between January 2011 and early February 2022, and recommends where management should act. The dataset holds 27,891 encounters, 974 patients and 47,701 procedures. Every headline figure was checked against an independent calculation on the raw files.

Key findings

1. **Activity.** Yearly encounters sat between 2,228 and 2,519 from 2015 to 2020, with two surges: 3,885 in 2014 and 3,530 in 2021. The 220 encounters shown for 2022 cover only five weeks and are not a decline.
2. **Mix.** Ambulatory (44.95%) and outpatient (22.59%) care make up about two thirds of encounters. Inpatient care is only 4.07%.
3. **Cost.** Total claim cost is 101.51 million, an average of 3,639.68 per encounter. Inpatient encounters are the most expensive (average 7,761) but account for 8.7% of cost. Urgent care is 13.1% of encounters yet 23.0% of cost, at an average of 6,369, about 2.2 times the ambulatory average of 2,894.
4. **Coverage.** Payers cover only 30.63% of total claim cost, which leaves 70.4 million (69.4%) uncovered. Uninsured encounters number 8,807 (31.6% of all encounters) and carry 49.3 million of claims, 48.5% of total cost. A further 4,779 encounters belong to a named insurer that recorded zero coverage, worth 13.8 million, 13.6% of cost. Among insured encounters the payer share rises to 59.5%.
5. **Duration.** The average encounter lasts 7.27 hours, but the median is 15 minutes. Only 75 encounters (0.27%) exceed 24 hours, 58 of them inpatient, yet a few run for months or years and distort the average.
6. **Procedures.** The 47,701 procedures span 163 types, and only 52.6% of encounters have any. The most frequent are routine screenings and assessments, led by health and social care needs assessment (4,596). The most expensive, such as hemodialysis (average about 29,300), are rare.

Recommendations

Priority	Recommendation	Evidence	Expected benefit
1	Audit the 4,779 insured encounters with zero coverage to separate denied claims from recording gaps, then	13.8 million of claims, 13.6% of total cost, sit with insurers that recorded nothing	Recovers or explains a large block of cost, and makes the payer coverage figure

Priority	Recommendation	Evidence	Expected benefit
	correct the claim process		trustworthy
2	Screen uninsured patients for eligibility for existing coverage programmes and financial assistance	8,807 uninsured encounters carry 49.3 million, 48.5% of cost, at an average of 5,593 each; Medicaid encounters almost never show zero coverage	Moves part of the uninsured burden onto payers and reduces unpaid exposure
3	Review the drivers of urgent care cost, including charge patterns and the mix of procedures	Urgent care takes 23.0% of cost from 13.1% of encounters	Targets the class where cost is out of proportion to volume
4	Validate the 75 long-duration records, close any open episodes, and report median duration beside the mean	13 encounters exceed 30 days, 4 exceed a year, and the longest is about 5 years	Gives reliable length-of-stay figures for capacity planning
5	Make reason codes and claim cost mandatory at data entry	70.1% of encounters have no reason recorded, and 89 have a zero claim cost	Allows analysis by clinical reason and removes avoidable cost errors

Caveats

- The data covers one hospital and an adult-only population, so results should not be generalised to other hospitals or to children.
- Zero coverage for an insured patient is unexplained in the data. It may reflect denials, pending claims or missing records.
- Uncovered cost is the claim amount that payers did not pay. It is not the same as lost revenue, because the hospital may have recovered it from patients.
- 2022 is a partial year and all 2022 figures are provisional.
- 89 zero-cost and 75 long-duration encounters were kept and flagged, not removed. Details are in the Data Quality Assessment and KPI Definition Sheet.

Suggested next steps

1. Investigate what drove the surges in 2014 and 2021 before using past volumes to forecast demand.
2. Obtain the full year of 2022 data, and repeat the coverage analysis once it is available.
3. Add a payer-level view of denied or pending claims if that data exists, to test recommendation 1 directly.

Limits and information needed for each recommendation

1. **Audit insured encounters with zero coverage.** The data does not say why coverage is zero, so denials, pending claims and missing records cannot be told apart. Needed: claim status and denial codes from the billing system.
2. **Screen uninsured patients for eligibility.** The data holds no income or eligibility information, and Medicaid's low zero-coverage rate does not prove the uninsured would qualify. Needed: eligibility rules and patient enrolment or income data.
3. **Review urgent care cost drivers.** The figures show claim cost by encounter class, not what makes urgent care expensive. Needed: itemised charges and the procedure mix for urgent care encounters.
4. **Validate long-duration records.** The data cannot separate genuine long episodes from recording errors. Needed: admission and discharge records from the source system.
5. **Make reason codes and claim cost mandatory.** Some blanks may be legitimate, because a routine wellness visit may have no reason to record. Needed: the data owner's entry rules, to learn which blanks are errors.